

Exercise Solutions: While Loop

1. Countdown from a Given Number

Solution

```
import java.util.Scanner;

public class Countdown {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Input starting number: ");
        int number = input.nextInt();

        while (number >= 1) {
            System.out.println(number);
            number--;
        }

        input.close();
    }
}
```

Sample Output

```
5
4
3
2
1
```

2. Sum of Digits of an Integer

Solution

```
import java.util.Scanner;

public class SumOfDigits {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Input number: ");
        int number = input.nextInt();
        int sum = 0;

        while (number > 0) {
            int digit = number % 10;
            sum += digit;
            number /= 10;
        }

        System.out.println("The sum of the digits is: " + sum);
        input.close();
    }
}
```

Sample Output

```
Input number: 5834
The sum of the digits is: 20
```

3. Reverse an Integer

Solution

```
import java.util.Scanner;

public class ReverseInteger {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Input number: ");
        int number = input.nextInt();
        int reversed = 0;

        while (number != 0) {
            int digit = number % 10;
            reversed = reversed * 10 + digit;
            number /= 10;
        }

        System.out.println("The reversed number is: " + reversed);
        input.close();
    }
}
```

Sample Output

```
Input number: 12345
The reversed number is: 54321
```

4. Check a Palindrome Number

Solution

```
import java.util.Scanner;

public class PalindromeNumber {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Input number: ");
        int number = input.nextInt();
        int original = number;
        int reversed = 0;

        while (number > 0) {
            int digit = number % 10;
            reversed = reversed * 10 + digit;
            number /= 10;
        }

        if (original == reversed) {
            System.out.println(original + " is a palindrome number");
        } else {
            System.out.println(original + " is not a palindrome number");
        }

        input.close();
    }
}
```

Sample Output

```
Input number: 1221
1221 is a palindrome number
```

5. Display a Fibonacci Sequence

Solution

```
import java.util.Scanner;

public class FibonacciSequence {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Input number of terms: ");
        int terms = input.nextInt();
        int first = 0;
        int second = 1;
        int count = 1;

        System.out.println("The Fibonacci sequence is:");
        while (count <= terms) {
            System.out.print(first);
            if (count < terms) {
                System.out.print(" ");
            }
            int next = first + second;
            first = second;
            second = next;
            count++;
        }
        System.out.println();

        input.close();
    }
}
```

Sample Output

```
Input number of terms: 8
The Fibonacci sequence is:
0 1 1 2 3 5 8 13
```